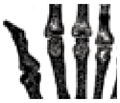


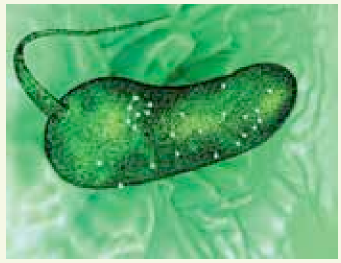
Elbows used to determine sex
Researchers have accurately guessed the sex of skeletons by examining the elbow (distal humerus) bone. **https://dgit.in/nov19-33**

The sensitive side of male spiders
A study revealed that male spiders may have a higher sensory capacity when they mate than previously thought. **https://dgit.in/nov19-34**

conditions during which water transitions from vapour to liquid to ice. Such a phase diagram allows researchers to understand why the organelles are formed in certain local regions within the cytoplasm of cells, and not in others. The studies can also help prevent any problems in the formation of MLOs. If the MLOs lose their consistency within cells, it can lead health problems including cancer, Alzheimer's disease and amyotrophic lateral sclerosis (ALS).

Using CasDrop, the researchers observed how the MLOs were formed within the nucleus of the cell. The MLOs would deform the chromatin in unexpected ways, selectively incorporating some proteins into their structures, and at the same time rejecting other proteins. Through this process, the chaotic liquid matter within the cells turn into functioning MLOs.

IMAGING A CELL IN THREE DIMENSIONS



Credit: **Kelly Lab / Penn State**

Researchers have been using a technique known as liquid-cell electron microscopy to study living cells, but so far the method has only produced 2D images. A team of researchers from Penn State have developed a new method called Liquid-Cell Electron Tomography (LCET) to observe the processes within living cells in 3D. The sample is slightly tilted for multiple scans, and then all the images are combined together or stacked using a 3D imaging software. Think of it as a CAT scan for a single cell. Shown here are pathogenic phages attacking a bacteria.

SEPARATION

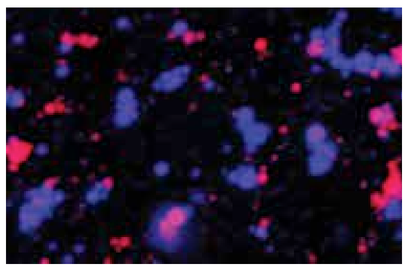
A puzzling aspect about MLOs was why they simply would not clump together to form larger liquid droplets. The MLOs are somehow able to maintain their structures despite being liquid bodies suspended within a liquid. Scientists are only now beginning to understand why.

Researchers from the University of Buffalo have discovered at least a partial explanation. The research indicates that the structure of the RNA and proteins that form the MLOs are at least partially responsible for them retaining their structures, and not mixing with other MLOs. Certain types of RNA and proteins are stickier than others, and while they clump together, they do not easily mix with other clumps. This tendency is shown by RNA that is rich in an aromatic compound called purine and proteins that are rich in an amino acid called arginine. The presence of these constituents cause the organelles to phase into a gel-like state, which prevents them from mixing. While the research did uncover a particular mechanism that causes the organelles to take on a gel-like state, there may be other reasons that have the same end result. It may not be that the presence of purine in RNA and the presence of arginine in proteins is the only reason for organelles to transform into the gel-like state, preventing them from mixing. Priya Banerjee, lead author of the paper says, "Cells are enormously complex, with many different molecules undergoing different processes that come together at the same time to influence what goes on inside MLOs. By using model systems, we are able to better understand how one particular variable may impact the formation and dissolution of these organelles. And we do expect to see these same forces at play in nature, inside cells."

THE NUCLEOLUS

Within the nucleus of the cell, is the nucleolus, an organelle within an organelle. It can be seen as a distinct dark spot. The nucleolus was first imaged in the 1830s, but little was known about it until 1964, when studies showed that frog eggs without the nucleolus were not capable of forming life. It was thus assumed that the nucleolus is necessary for life. Subsequent studies showed that the nucleolus served as the centre for cellular growth, within the "brain" or nucleus of the cell.

The nucleolus is made up of liquid, roughly the consistency of honey, but it has a sophisticated internal structure as well. While it does have a clear boundary, the nucleolus does not have a membrane. It was not clear how the liquids that form the nucleolus could maintain their separation, and the internal structure. Researchers from Princeton have uncovered that the RNA chains and proteins within the nucleoli spontaneously assemble into three distinct liquid layers. To conduct the study, the researchers used frog eggs, which have a large, distinct nucleolus. These liquids have different properties including surface tension and viscosity. It is a difference in these properties that allow them to form the



Credit: **Ibraheem Alshareedah / Priya Banerjee Laboratory**

The blue and pink droplets are both composed of a mixture of protein and RNA, but do not mix. They are tagged here with dyes for easy identification.

complex structures within the nucleolus. The separation is comparable to how water and oil can co-exist without mixing, despite both being liquids. The study showed that two nucleolar proteins, called FIB1 and NPM1, would not mix together as a homogeneous liquid. Nilesh Vaidya, co-lead author of the study says, "We've provided a biophysical mechanism for the structure of the nucleolus that automatically emerges from the collective behaviour of immiscible liquids."